ARIADNE

Mapping the way back

Observe the space. Verify connections.
Choose the next view. Preserve the way home.

COMPONENT TAGS

- CV — OpenCV 5 modules
- AWS — AWS-hosted service
- CL — COOL integration
- AV — Agentic vision chooser

COMPOSITE PIPELINE

→ data / geometry ⇢ feedback / authority

START READING HERE

SELECTIVE COLLABORATION / ASYNCHRONOUS WORKERS

RF attenuation / partitions · broker election + epochs · ARQ / buffering / resync · exact byte ledger

CV **Recognize overlap**
EigenPlaces + cosine index
Global descriptor retrieval

candidates

Spend bytes on evidence
Greedy / spectral λ_2
Connectivity gain / estimated byte

selected pairs

CV **Verify shared geometry**
ALIKED + LightGlue / PnP
Degeneracy + known-entry gate

keyframe cues

relative SE(3) + covariance

IMU / preintegration

PER-AGENT ESTIMATION / OPENCV 5 + EIGEN

CV CL **Photometric resilience**
Flat-field / VST
Depth-aware falloff
Exposure control

CV CL **Visual front end**
Rectify / LK + ORB
SGBM / σ_Z / landmarks
Keyframe evidence

CV **Stereo-inertial SLAM**
IMU factors / SE(3) window BA
Schur LM / local graph
Pose + covariance

World + sensor testbed
Procedural maze / hazards
ROS 2 + Gazebo Harmonic
Stereo RGB / IMU / lamp / movers

exposure / lamp commands

motion changes observations

keyframe pose / evidence

stereo depth / uncertainty

Robust shared map
PCM → chordal → GNC-TLS
Known-entry / alias gating
Versioned poses / incomplete status

global alignment& versions

SPATIAL OUTPUTS

AWS **Gaussian reconstruction**
RGB / depth / pose keyframe tee
Anchored submaps / gsplat
Versioned placements / inspection

clearance

Responder egress
Multi-exit D* Lite
Clearance / support / risk

weak support

path + support

AWS **Incident-command console**
Spark / WebGL scene + route
Fleet / bytes / operator controls

CLOSED-LOOP MISSION CONTROL

Validate + execute
Feasibility / risk / approval
goto / follow / hold

choice

AV AWS **Bounded agent decisions**
MCP tools + Bedrock
Candidate choice / rationale

feasible tasks

Explore + allocate
NBV: coverage / uncertainty
Route support / costs + CBBA

frontiers / route gaps

Geometric world model
TSDF → agent / responder ESDF
Frontiers / capsules / support / risk

map / uncertainty / route / fleet

operator intent / directives

approval / rejection (dispatch gate)

RECORD + EVALUATE MCAP / deterministic replay | EuRoC / TUM / Hilti / seeded scenes
Accuracy, bytes, coverage, route validity | independent oracles, manifests, replay digests

AWS CL **AWS TESTBED** Terraform / IAM | ECR | EC2 Spot CPU / GPU / Arm | Bedrock | hosted judging frontend / live endpoint | S3

CONTRACTS SE(3) + covariance | frame-tagged keyframes | FlatBuffers / bytes | seeded replay + provenance